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Nanomedicine & Lipid Nanoparticle (LNP) Formulation | Physicochemical Characterization | Biomaterials & Delivery Systems

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Research Summary: Structural Architecture of Polymer-Lipid Nanodiscs

The extraction of membrane proteins from native cellular environments frequently induces conformational denaturation as a result of lipid bilayer disruption. Amphipathic polymers, including vinyl ether–maleic anhydride (VEMA) and styrene–maleic acid (SMA) copolymers, circumvent this instability by circumscribing discrete segments of the lipid bilayer to yield soluble, stable nanodiscs. These polymer-bounded nanodisc assemblies exhibit tunable dimensions and structural stability across diverse pH regimes and divalent cation concentrations.

The schematic summary below illustrates the structural organization and four-pillar development of this molecular assembly (*Shah et al., Biomacromolecules, 2024, doi: 10.1021/acs.biomac.4c00772*):

1. Membrane Extraction & Lipid Nanodisc Self-Assembly: Amphipathic VEMA copolymers (with butyl or octyl aliphatic side chains, R = C_4H_9 or C_8H_{17}) spontaneously partition into phospholipid bilayers (POPC/POPG) to form uniform, water-soluble polymer-bounded lipid nanodiscs (~10 nm diameter by DLS and TEM, extending from 3 nm to 60 nm via block copolymer engineering) without requiring detergents.

2. Controlled RAFT Polymer Synthesis & Chemistry: Alternating copolymerization of alkyl vinyl ethers and maleic anhydride at 60–80 °C using AIBN/LPO initiators and four screened RAFT chain transfer agents (EtPAX, PAETC, PADTC, iBADTC), followed by mild aqueous NaOH hydrolysis at 50 °C or nucleophilic ring-opening functionalization (PEGylation, amino acids, and 9 cationic/anionic/zwitterionic VEMADs), verified via 1D ^{1}H/DOSY NMR and SEC/GPC (Đ ~ 1.3–1.5, Mn = 6–40 kDa).

3. Biophysical Characterization & Native Protein Performance: Reconstitution and stabilization of sensitive membrane proteins (such as KCNE1, KCNE3, KCNE4, gp28, and human TRPV1) in native-like lipid bilayers, validated by site-directed spin labeling Continuous-Wave Electron Paramagnetic Resonance (CW-EPR) spectra confirming preserved native conformational dynamics compared to vesicles.

4. Superior Physiological Stability & Compatibility (VEMA vs. SMA): Turbidimetric (O.D. 620 nm) stability profiles demonstrating that aliphatic VEMA nanodiscs remain optically clear and resist aggregation across broad pH regimes (**pH 2.0–10.0**) and high divalent cation concentrations (**up to 30–100+ mM Mg^{2+}**), overcoming the severe acidic precipitation and Mg^{2+} chelation sensitivity of aromatic SMA polymers.

(Full graphical schematic available in the Visual Summary: Polymer-Lipid Nanodisc Architecture document)

EXECUTIVE SUMMARY

Results-driven **Nanomedicine, Biomaterials, and Formulation Scientist** with **5+ years of research experience** engineering self-assembled **lipid nanoparticles (LNPs)**, lipid–polymer nanodiscs, amphiphilic block copolymers, and charge-tunable macromolecular carriers for biological encapsulation and delivery. Proven track record of scientific ownership and preliminary data generation across **9 peer-reviewed publications** (*Biomacromolecules*, *ACS Macro Letters*, *Polymer Chemistry*, *Polymer*, *BBA-Biomembranes*)

and **6 manuscripts under review/in preparation**. Expert in **lipid nanoparticle formulation**, **surface modification (PEGylation, amino acid, and cationic/anionic/zwitterionic charge tuning)**, **comprehensive physicochemical characterization (DLS, TEM/SEM, SEC/GPC, HPLC, 1D ^{1}H/DOSY NMR, CW-EPR, UV-Vis, FTIR)**, and **physiological stability and quality control assays (pH 3.0–9.0, divalent cation tolerance)**. Experienced mentor (**13 trainees supervised**) adept at maintaining rigorous electronic laboratory records, training new lab members in established protocols, and bridging nanoparticle physicochemical properties with biological performance across multidisciplinary teams.

EDUCATION

- **Ph.D. in Chemistry (Polymer Science, Materials Chemistry & Biomaterials)** | Miami University, Oxford, OH | **Oct 2026**
    - *Dissertation:* "Vinyl Ether-Maleic Anhydride Copolymers Facilitating Native-Like Lipid Membrane Environments"
    - *Advisors:* Prof. Dominik Konkolewicz & Prof. Rick Page (in collaboration with Prof. Gary Lorigan)
- **M.S. in Material Chemistry** | Miami University, Oxford, OH | **2023**
- **M.S. in Physical Chemistry** | Government College University, Faisalabad | **2017**

CORE COMPETENCIES

- **Lipid Nanoparticle (LNP) Formulation & Surface Engineering:** Self-Assembled Lipid–Polymer Nanoparticles (POPC/POPG, SMALPs, VEMA Nanodiscs), Scalable Thin-Film & Rapid Aqueous Formulation, Post-Polymerization Nucleophilic Conjugation (**PEGylation**, Amino Acid & Cationic/Anionic/Zwitterionic Charge Tuning), Controlled RAFT Polymerization, Amphiphilic Block Copolymer Engineering.
- **Physicochemical Characterization & Quality Control:** Dynamic Light Scattering (**DLS** – Hydrodynamic Size & Polydispersity), Transmission & Scanning Electron Microscopy (**TEM, SEM** – Nanoparticle Morphology), High-Performance Liquid Chromatography (**HPLC**), Size Exclusion / Gel Permeation Chromatography (**SEC/GPC**), Polyelectrolyte & Surface Charge Titration, 1D ^{1}H/DOSY Nuclear Magnetic Resonance (**NMR**), Continuous-Wave Electron Paramagnetic Resonance (**CW-EPR**), **UV-Vis**, **FTIR**, **DSC**, Rheometry, Instron Mechanical Testing, **ICP-OES**, **AAS**.
- **Colloidal Stability, Biologics Encapsulation & Assay Development:** Physiological Stability Testing (**pH 3.0–9.0**, up to **30 mM Mg^{2+}** Divalent Ion Tolerance), Solubilization & Encapsulation Kinetics, Reconstitution & Stabilization of Biological Targets in Lipid Bilayers (**Human TRPV1**, **KCNE1**, **KCNE4**, **gp28**), Biomimetic Vesicle Remodeling, Antiviral Bilayer-Disrupting Networks.
- **Lab Management, Documentation & Mentorship:** Preliminary Data Generation for NIH/NSF Grant Proposals, Electronic Lab Notebook (ELN) & Data Storage Management, Protocol Standardization & Onboarding Training (11 Undergraduates & 2 Graduate Students Mentored), Multidisciplinary Collaboration.
- **Computational & Design Tools:** KaleidaGraph, Bruker TopSpin, Fiji/ImageJ, Autodesk Fusion 360 (CAD & 3D Printing Fabrication), Adobe Illustrator, Python (Kinetic Modeling & Data Visualization basics), MATLAB-basics.

RESEARCH EXPERIENCE

Graduate Research Fellow – Lipid Nanoparticle Formulation, Biomaterials & Physicochemical Characterization

Miami University | *Laboratories of Prof. Dominik Konkolewicz, Prof. Gary Lorigan & Prof. Rick Page (NIH- & NSF-Funded)* | **May 2021 – Present**

- **Formulation & Size Control of Self-Assembled Lipid Nanoparticles:** Spearheaded the rational design and kinetic modeling of alternating and block **vinyl ether–maleic anhydride (VEMA)** copolymers via RAFT polymerization; screened 4 CTAs (iBADTC, PADTC, EtPAX, PAETC) to achieve low dispersity (Đ ~ 1.3–1.5) and programmable molecular weights (Mn = 6–40 kDa) (*Biomacromolecules* 2024; *Polymer* 2026). Chain-extended VEMA macro-CTAs with hydrophobic *n*-butyl acrylate (nBA) and *tert*-butyl acrylate (tBA) followed by selective hydrolysis to acrylic acid (AA) blocks (*Polymer Chemistry* 2026), systematically tuning self-assembled lipid nanoparticle diameter from **60 nm down to 3 nm**.
- **PEGylation, Cationic/Anionic Charge Tuning & Physiological Quality Control:** Exploited the reactive maleic anhydride backbone for modular nucleophilic ring-opening conjugation with **polyethylene glycol (PEG)** chains, **amino acids**, and charged modifiers to generate **9 charge-tunable derivatives (VEMADs)**. Formulated cationic (BVEMA-DP28-Pos) and anionic (MVEMA-AE) lipid nanoparticles that resist aggregation and matrix degradation across broad physiological conditions (**pH 3.0–9.0** and **up to 30 mM Mg^{2+}**).
- **Rapid, Gentle Aqueous Formulation for Sensitive Biologics:** Synthesized aliphatic **cyclohexyl-VEMA (CH-VEMA)** copolymers (6 kDa and 12 kDa) that achieve **spontaneous aqueous lipid self-assembly in under 60 seconds** without mechanical shear, organic solvents, or freeze–thaw cycles, maintaining ~10 nm colloidal stability for over 72 hours. Co-developed scalable substrate thin-film fabrication protocols for lipid nanoparticles (*Analytical Biochemistry* 2022) and investigated concentration-dependent POPC lipid vesicle remodeling (*ACS Omega* submitted).
- **Multi-Scale Physicochemical Characterization & Biological Target Encapsulation:** Integrated **DLS, TEM, SEM, SEC/GPC, HPLC, UV-Vis, FTIR, and 1D ^{1}H/DOSY NMR** with site-directed spin labeling **CW-EPR spectroscopy** to quantify nanoparticle size distribution, lipid saturation effects, local microviscosity, and bilayer packing (*BBA-Biomembranes* 2025; *Biophysical Chemistry* 2025; *Chemistry and Physics of Lipids* 2025). Using purified proteins provided by collaborators in the Lorigan lab, reconstituted and encapsulated **human TRPV1**, **KCNE1**, **KCNE4**, and **gp28** into POPC/POPG lipid nanodiscs while preserving native conformational states.
- **Functional Polymer Networks, Sorbents & Grant Data Generation:** Synthesized cross-linked polymer networks incorporating lipid-bilayer disrupting macromonomers for antiviral applications (*Polymer Chemistry* 2022) and reusable high-capacity chelation networks (*ACS Macro Letters* 2026), while generating key preliminary datasets supporting federal grant renewals and training 13 junior researchers in established laboratory protocols.

Graduate Research Assistant – Polymer Chemistry & Advanced Oxidation Processes

Government College University Faisalabad | *Laboratory of Dr. Munir* | **2015 – 2017**

- Investigated the radiolytic degradation kinetics of organic compounds using **Gamma Irradiation** and evaluated amphiphilic copolymer formulations for aqueous contaminant sequestration.

PEER-REVIEWED PUBLICATIONS

Published & In-Press Journal Articles

1. **Shah, M.Z.**, Rotich, N.C., Okorafor, E.A., Oestreicher, Z., Demidovich, G., Eapen, J., Page, R.C., Konkolewicz, D., & Lorigan, G.A. (2024). Vinyl ether maleic acid polymers: tunable polymers for self-assembled lipid nanodiscs and environments for membrane proteins. ***Biomacromolecules***, 25(10), 6611–6622.
2. **Shah, M.Z.**, Okorafor, E., Rotich, N.C., Henoch, Q., Thapa Acharya, R., Page, R.C., Lorigan, G.A., & Konkolewicz, D. (2026). Vinyl ether maleic acid block copolymers: a versatile platform for tunable self-assembled lipid nanodiscs and membrane protein characterization. ***Polymer Chemistry***, 17(2), 194–206.
3. Acharya, R.T., **Shah, M.Z.**, Woodruff, A., & Konkolewicz, D. (2026). Vinyl Ether Maleic Anhydride Copolymers: Efficient and Reusable Sorbents for Removing Heavy Metals from Water. ***ACS Macro Letters***, 15(7), 1035–1041.
4. Acharya, R.T., **Shah, M.Z.**, Woodruff, A., & Konkolewicz, D. (2026). Kinetics of RAFT Copolymerization of Vinyl Ether and Maleic Anhydride Monomers. ***Polymer***, 129999.
5. Okorafor, E.A., Gordon, E.A., Sahu, I.D., **Shah, M.Z.**, Konkolewicz, D., & Lorigan, G.A. (2025). Influence of lipid saturation on the structural properties of styrene maleic acid lipid nanoparticles (SMALPs). ***Biochimica et Biophysica Acta (BBA) - Biomembranes***, 1867(5-6), 184424.
6. Gordon, E.A., Okorafor, E.A., Sahu, I.D., Burridge, K.M., **Shah, M.Z.**, Thapa, O., Konkolewicz, D., & Lorigan, G.A. (2025). The effect of lipid saturation on the formation of styrene maleic acid lipid nanoparticles. ***Biophysical Chemistry***, 107566.
7. Rotich, N.C., Okorafor, E.A., Sahu, I.D., **Shah, M.Z.**, Konkolewicz, D., & Lorigan, G.A. (2025). A comparison of the effect of SMA derivatives on the structural topology and dynamics of two bacteriophage peptides. ***Chemistry and Physics of Lipids***, 105562.
8. Gordon, E.A., Richardson, Y.B., **Shah, M.Z.**, Burridge, K.M., Konkolewicz, D., & Lorigan, G.A. (2022). Formation of styrene maleic acid lipid nanoparticles (SMALPs) using SMA thin film on a substrate. ***Analytical Biochemistry***, 647, 114692.
9. Burridge, K.M., Rahman, M.S., De Alwis Watuthanthrige, N., Gordon, E., **Shah, M.Z.**, et al. (2022). Network polymers incorporating lipid-bilayer disrupting polymers: towards antiviral functionality. ***Polymer Chemistry***, 13(31), 4547–4556.

SCIENTIFIC LEADERSHIP, MENTORSHIP & TEACHING

- **Research Mentor (11 Undergraduates & 2 Graduate Students)** | Miami University (**2022 – Present**)
    - Directly trained and supervised 13 researchers in lipid nanoparticle formulation, RAFT synthesis, PEG/amino-acid polymer derivatization, NMR/GPC/DLS quality control, and electronic record-keeping, resulting in **5+ co-authored manuscripts** (including Cubberly et al. and Hicov et al., 2026).
- **President** | American Chemical Society (ACS) POLY/PMSE Cincinnati Chapter (**2024 – 2025**)
    - Spearheaded cross-institutional symposia, poster judging, and networking initiatives connecting academic polymer scientists with industrial and biomedical leaders; previously served as **Treasurer (2022 – 2023)**.
- **Graduate Teaching Assistant** | Department of Chemistry & Biochemistry, Miami University (**2021 – 2023**)
    - Instructed laboratory and recitation sections for **Organic Chemistry I & II** (CHM 244/245) and **General Chemistry I & II** (CHM 144/145).
- **Assistant Education Officer** | School Education Department, Punjab (**2017 – 2020**)
    - Supervised curriculum execution, STEM pedagogy, and institutional administration across public and private schools.
- **Peer Reviewer** | *Journal on Excellence in College Teaching* (**2023 – Present**)

HONORS, FELLOWSHIPS & AWARDS

- **Poster Judge & PMSE Division Representative** | ACS Fall 2025 National Meeting, Washington, DC (Aug 2025)
- **Best Presenter Award** | American Chemical Society (ACS) Southwest Ohio Chapter (Oct 2024)
- **Graduate Research Fellowship (2x Recipient)** | Miami University (May 2022 & May 2023)
- **Competitive Graduate Travel Awards (3x Recipient)** | ACS Fall 2025 (Washington, DC), SMALPs Conference 2023 (Knoxville, TN), and ACS Spring 2023 (Indianapolis, IN)

SELECTED NATIONAL & INTERNATIONAL PRESENTATIONS

1. **Shah, M.Z.** & Konkolewicz, D. *Vinyl Ethers–Maleic Anhydride Lipid Nanodiscs: Synthesis, Block Extension, and Biophysical Performance.* (Dual Poster Presentations, **PMSE & POLY Divisions**), **ACS Fall 2025 National Meeting**, Washington, DC (Aug 2025).
2. **Shah, M.Z.** *Polymer Systems for Studying Membrane Proteins in Native-Like Environments.* (Invited Oral Presentation), **International Conference on Functional Materials and Emerging Technologies**, Faisalabad (Nov 2024).
3. **Shah, M.Z.** *Polymer-Based Lipid Nanodiscs: Tunable Macromolecular Design.* (Oral Presentation), **Graduate Research Forum**, Miami University, OH (Nov 2024).
4. **Shah, M.Z.** *Vinyl Ethers–Maleic Anhydride Lipid Nanodiscs.* (Award-Winning Oral Presentation), **ACS Southwest Ohio Chapter Symposium** (Oct 2024).

5. **Shah, M.Z.** *Tunable VEMA Copolymers for Membrane Protein Extraction.* (Oral & Poster Presentations), **SMALPs Conference**, University of Tennessee, Knoxville, TN (Nov 2023).
6. **Shah, M.Z.** *Macromolecular Engineering of Amphiphilic Copolymers.* (Flash Talk), **ACS POLY/PMSE Symposium**, University of Cincinnati, OH (July 2023).
7. **Shah, M.Z.** *Kinetic Study of Butyl Vinyl Ether and Maleic Anhydride Copolymerization & Lipid Disc Formation.* (2 Oral Presentations), **ACS Spring 2023 National Meeting**, Indianapolis, IN (March 2023).